Abhishek Lakhani

Applied AI Engineer – NLP & LLM Systems | M.Sc. Student

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Work authorisation: no visa sponsorship required | English (fluent), German (B1)

PROFILE

M.Sc. student in Automotive Software Engineering at TU Chemnitz who builds and measures NLP and LLM systems in Python. I launched SmartVisual, a live natural-language data analytics app open to the public, and built tested projects covering hybrid RAG with retrieval benchmarking (nDCG@10 0.728 vs 0.567 for TF-IDF), LLM structured extraction with Pydantic validation and a fine-tuned German DistilBERT classifier (0.897 macro-F1) deployed with ONNX int8 quantization (1.46x throughput, 4x smaller). Practical experience from a machine learning internship (federated learning, TensorFlow) and two software internships. Looking for working-student, thesis or junior roles in Applied AI and ML engineering.

SKILLS

Programming: Python, SQL, JavaScript, TypeScript, HTML/CSS

LLM & NLP: LLMs, Prompt Engineering, Few-Shot Prompting, Structured Outputs, RAG, LangChain, Hugging Face Transformers, Fine-Tuning (DistilBERT), Sentence Embeddings, Text Classification, Information Retrieval, LLM APIs (Anthropic, Ollama)

Search & Vector DB: Qdrant, FAISS, BM25, TF-IDF, Hybrid Search (RRF), Cross-Encoder Reranking

Machine Learning: PyTorch, TensorFlow, scikit-learn, pandas, NumPy, Model Evaluation (nDCG, MRR, Recall@k, F1, Significance Tests)

MLOps & Backend: FastAPI, REST APIs, Pydantic, Docker, Docker Compose, ONNX Runtime, Quantization, pytest, GitHub Actions (CI), SQLAlchemy, PostgreSQL, Streamlit

Web & Deployment: React, Next.js, Tailwind CSS, Supabase, Prisma, Vitest, Playwright, Cloudflare Workers, Vercel, Render

Tools & Methods: Git, GitHub, Linux, n8n, CVAT, Agile/Scrum, Code Reviews

PROFESSIONAL EXPERIENCE

Machine Learning Intern – Technische Universität Chemnitz, Chemnitz, Germany | 06/2025 – 01/2026

- Designed a privacy-focused image classification pipeline in TensorFlow and ran federated learning experiments across decentralised Solid Pod data sources.
- Built modular Python components (structured logging, custom exceptions, CSV-based result tracking) so that experiments could be reproduced and compared.

Software Development Intern – Wokontech IT Solution, Surat, India | 10/2021 – 03/2022

- Built and trained object detection pipelines on a dataset of 5,000+ images and improved model robustness through data curation.
- Evaluated models with standard detection metrics and ran the annotation workflow in CVAT.

Backend Development Intern – White Orange Software, Surat, India | 12/2020 – 05/2021

- Built and maintained backend features in Core PHP and tuned MySQL queries for performance.
- Took part in code reviews and wrote technical documentation in a cross-functional team.

PROJECTS

SmartVisual – AI Data Analytics App (own product, live) | React, TypeScript, Supabase (Postgres, Edge Functions), LLM API, Cloudflare Workers, GitHub Actions

Live at smartvisual.eu: anyone can sign up and use it (code private, walkthrough on request)

- Built and launched a web app where users upload CSV/Excel files and ask questions in English or German to get KPIs, 18 chart types, maps and short written insights.
- Designed for correct numbers: the LLM only interprets the question and phrases the explanation, while every figure is computed in code over the full dataset; a 22-question evaluation set on 3 datasets runs in CI (every value within 0.01 of independently computed answers).

RAG Document Q&A Service + Retrieval Benchmark | Python, FastAPI, BM25, BGE Embeddings, FAISS, OCR, SSE, Docker

github.com/abhisheklakhani-it/rag-docqa | github.com/abhisheklakhani-it/retrieval-eval-bench

- Built a retrieval-augmented generation (RAG) service that answers questions on uploaded PDFs, including scanned PDFs via OCR, with cited answers streamed over Server-Sent Events.
- Benchmarked TF-IDF, BM25, dense and hybrid retrieval on BEIR SciFact with a train/test split and significance tests, then switched the service to hybrid BM25 + dense retrieval: nDCG@10 0.728 vs 0.567 for TF-IDF. A calibrated similarity floor rejects 87% of off-topic questions; 73 tests.

German Text Classifier + ONNX Deployment | Hugging Face Transformers, PyTorch, Qdrant, ONNX Runtime, int8 Quantization, FastAPI, Docker

github.com/abhisheklakhani-it/german-text-classifier | github.com/abhisheklakhani-it/onnx-inference-optimization

- Fine-tuned distilbert-base-german-cased on 10kGNAD news (9 topics): 0.897 macro-F1 versus 0.845 for an embeddings + Qdrant kNN baseline; McNemar test $p < 0.001$.
- Exported the model to ONNX with graph fusion and int8 dynamic quantization: 1.46x PyTorch throughput, 4x smaller model (68 MB) for -0.3 points accuracy, served by FastAPI from a 147 MB Docker image.

LLM Structured Extractor | LLM Structured Outputs, Pydantic v2, FastAPI

github.com/abhisheklakhani-it/llm-structured-extractor

- Extracts validated JSON (title, company, skills, salary, remote policy) from messy English and German job postings, with Pydantic schema validation and an automatic retry on validation errors.
- Evaluated field by field on a held-out set; error-driven prompt changes raised all-fields-correct from 0.40 to 0.67 (mean of 3 runs). Served by FastAPI, 43 automated tests.

More projects at github.com/abhisheklakhani-it: Semantic Search API (FastAPI, Qdrant hybrid search, nDCG@10 0.737), Test Bench Commissioning Web App (FastAPI, PostgreSQL, 137 tests, live demo), LinkedIn Post Generator (LangChain, 34 tests). Private, code on request: Event Staff Scheduling Platform (Next.js, PostgreSQL, 330+ tests).

EDUCATION

M.Sc. Automotive Software Engineering – Technische Universität Chemnitz, Chemnitz, Germany | 04/2022 – present

- Master's thesis (in preparation): Real-time LiDAR health monitoring system with GUI for railway applications.

B.E. Computer Science and Engineering – Gujarat Technological University (Government Engineering College Dahod), Gujarat, India | 08/2017 – 07/2021

LANGUAGES

English – fluent | German – B1 (improving)

VOLUNTEERING

- Erasmus+ Youth Exchange (international project on gender equality, France); volunteer at a public cycling event in Chemnitz.